

This note specifies the supply side that pairs with the panel nested-logit demand model, derives every object from the demand primitives, and gives explicit steps to (i) estimate marginal cost and (ii) solve counterfactuals. The recommended specification is a **homogeneous-good industry pricing relation** with the **competitive / constant-marginal-cost** case imposed; the general conduct-parameter model is carried through so the competitive case is visible as a restriction.

## 1. Environment and notation

Markets are state-month pairs  $(s, m)$ ; write the subscript  $ms$ . In each market there is a single **representative e-cigarette** sold at price  $p_{ms}$  — the mL-sales-weighted index  $p_{ms} = (\text{avg price/mL})_{ms} \times (\text{avg capacity})_{ms}$  from the demand model. Let

- $\tau_{ms}$  — the specific excise tax per representative e-cigarette (built with the same weights as  $p_{ms}$ ; "analogous formula for the tax instrument" in your slides);
- $c_{ms}$  — marginal cost per representative e-cigarette (net of tax), assumed constant in quantity within a market;
- $N_{ms}$  — the representative nicotine content (the mL-weighted  $N_{\text{hom}}$ );
- $M_{ms}$  — market size (number of agents), taken as given by the pricing decision.

The e-cigarette is produced by many firms/SKUs; we do **not** model them individually. Instead a scalar **conduct parameter**  $\theta \in [0, 1]$  indexes industry pricing, so a single homogeneous-good relation summarizes supply.

## 2. Demand elasticity of the representative e-cigarette

The pricing relation needs the own-price response of e-cigarette demand, which comes entirely from your estimated demand. Your nests are  $g \in \{\text{Not-smoke}, \text{Smoke}\}$  with  $\text{Smoke} = \{c, e\}$  (cigarette, e-cigarette) and nest scale  $\lambda \equiv \lambda_{\text{Smoke}} \in (0, 1]$  ( $\hat{\lambda} = 0.624$ ). With representative utility  $V_{j,ms} = \beta^p p_{j,ms} + \dots$  and common price coefficient  $\beta^p < 0$ , the choice probability factors as

$$s_{e,ms} = \underbrace{\frac{\exp(V_e/\lambda)}{\sum_{k \in \text{Smoke}} \exp(V_k/\lambda)}}_{P_{e|S}} \cdot \underbrace{\frac{\exp(IV_S)}{\sum_g \exp(IV_g)}}_{P_S}, \quad IV_S = \lambda \ln \sum_{k \in \text{Smoke}} \exp(V_k/\lambda).$$

**Derivation of the own-price semi-elasticity.** Because  $\partial V_e / \partial p_e = \beta^p$ ,

$$\frac{\partial \ln P_{e|S}}{\partial p_e} = \frac{\beta^p}{\lambda} (1 - P_{e|S}), \quad \frac{\partial IV_S}{\partial p_e} = \beta^p P_{e|S}, \quad \frac{\partial \ln P_S}{\partial p_e} = \beta^p P_{e|S} (1 - P_S).$$

Adding the two log-derivatives ( $\ln s_e = \ln P_{e|S} + \ln P_S$ ) gives the key object

$$D_{ms} \equiv \frac{\partial \ln s_{e,ms}}{\partial p_{ms}} = \beta^p \underbrace{\left[ \frac{1}{\lambda} (1 - P_{e|S}) + P_{e|S} (1 - P_S) \right]}_{A_{ms} > 0} < 0,$$

and the own-price **elasticity**  $\eta_{ms} = p_{ms} D_{ms} = \beta^p p_{ms} A_{ms}$ . All of  $P_{e|S}$ ,  $P_S$ ,  $\lambda$ ,  $\beta^p$  come from your estimates, so  $D_{ms}$  and  $\eta_{ms}$  are computed market-by-market with no new parameters. (As  $\lambda \rightarrow 1$  this collapses to simple logit,  $A = 1 - s_e$ ; smaller  $\lambda$  raises the within-nest term, i.e. cigarettes are the closest substitute.)

### 3. The pricing (supply) relation

Aggregate e-cigarette quantity is  $Q_{ms} = M_{ms} s_{e,ms}$ . With constant marginal cost and a specific tax remitted by sellers, the homogeneous-good conduct FOC is

$$p_{ms} - (c_{ms} + \tau_{ms}) = -\theta \frac{Q_{ms}}{\partial Q_{ms} / \partial p_{ms}} = -\frac{\theta}{D_{ms}} = \frac{\theta}{|\beta^p| A_{ms}} \geq 0.$$

$M_{ms}$  cancels; the markup depends only on demand objects and  $\theta$ . Special cases:

| Conduct                            | $\theta$ | Pricing relation                                                       |
|------------------------------------|----------|------------------------------------------------------------------------|
| Perfect competition / price-taking | 0        | $p_{ms} = c_{ms} + \tau_{ms}$                                          |
| Symmetric $n$ -firm Cournot        | $1/n$    | $p_{ms} = c_{ms} + \tau_{ms} + \frac{1}{n} \frac{1}{ \beta^p  A_{ms}}$ |
| Monopoly / perfect collusion       | 1        | $p_{ms} = c_{ms} + \tau_{ms} + \frac{1}{ \beta^p  A_{ms}}$             |

The cigarette side is symmetric, with its own  $A_{ms}^c$  (using  $P_{c|S}$ ) and  $\theta^c$ . Under  $\theta = 0$  the two sides decouple (each price equals its own cost+tax), which sidesteps cross-price strategic interactions between cigarettes and e-cigarettes even though they share a nest.

### 4. Marginal-cost specification

Model net-of-tax marginal cost as

$$c_{ms} = \mathbf{w}'_{ms} \gamma + \gamma_N N_{ms} + \phi_s + \psi_{\text{year}} + \omega_{ms},$$

where  $\mathbf{w}_{ms}$  are input-cost shifters (e.g. wholesale/PPI hardware and propylene-glycol/nicotine input proxies, import price indices),  $N_{ms}$  is the representative nicotine content (higher delivered nicotine is costlier and, post-2016, technologically gated by salts),  $\phi_s, \psi_{\text{year}}$  absorb region/time cost levels, and  $\omega_{ms}$  is the structural cost shock. Nicotine enters **cost**, not a firm choice: in your model nicotine acts dynamically through the stock  $N_{im}$  and the representative content is a market aggregate, so a nicotine FOC would require a differentiated-products setup you have avoided.

### 5. Pass-through, and why impose competition

Differentiate the pricing relation in  $\tau$ , noting  $A_{ms}$  depends on  $\tau$  only through  $p$ :

$$\rho_{ms} \equiv \frac{dp_{ms}}{d\tau_{ms}} = \frac{1}{1 - \frac{\theta}{|\beta^p|} \frac{d}{dp} \left( \frac{1}{A_{ms}} \right)}.$$

At  $\theta = 0$ ,  $\rho = 1$  (full pass-through) exactly. For  $\theta > 0$  pass-through departs from one by a term proportional to demand curvature  $\frac{d}{dp} (1/A)$ . Your own references estimate near-unit pass-through — Cotti et al. (2022) put e-cig pass-through at 0.90–1.01 and He et al. (2023) put cigarette pass-through at 1.00–1.10 — which is the empirical signature of  $\theta \approx 0$  with roughly constant marginal cost.

**Identification of conduct.** Separating  $\theta$  from  $c$  requires *demand rotators* (instruments that shift the slope of demand). Your only excluded price instrument is the excise tax, which you already use to instrument price in the demand second stage; a tax shifts cost and, through price, the demand level, but does not cleanly rotate the demand slope, so  $\theta$  is weakly identified (Corts 1999). Given (i) the pass-

through evidence and (ii) this identification limit, **impose**  $\theta = 0$  (or a single constant markup) rather than estimate  $\theta$ .

## 6. Estimation steps (recover the cost function)

1. **Demand objects.** From the estimated demand, compute  $P_{e|S,ms}$ ,  $P_{S,ms}$ ,  $s_{e,ms}$ , and  $A_{ms} = \frac{1}{\lambda}(1 - P_{e|S}) + P_{e|S}(1 - P_S)$ ; form  $D_{ms} = \beta^p A_{ms}$  and  $\eta_{ms} = p_{ms} D_{ms}$  at every  $(s, m)$ .
2. **Impose conduct.** Set  $\theta = 0$ : implied net-of-tax marginal cost is  $\hat{c}_{ms} = p_{ms} - \tau_{ms}$ . (Constant-markup variant:  $\hat{c}_{ms} = p_{ms} - \tau_{ms} - \mu$ , with  $\mu$  a single scalar estimated in step 3.)
3. **Cost regression.** Regress  $\hat{c}_{ms}$  on  $\mathbf{w}_{ms}$ ,  $N_{ms}$ , and fixed effects to recover  $\gamma$ ,  $\gamma_N$  and the residual  $\omega_{ms}$ . If instead you *estimate* a constant markup or  $\theta$ , do so by GMM using cost instruments  $\mathbf{w}_{ms}$  (and, for  $\theta$ , any credible demand rotator), stacking the demand and supply moments; bootstrap as in your demand step.
4. **Validate.** Compute model pass-through  $\rho_{ms}$  (Section 5) and check it against 0.90–1.10. Under imposed  $\theta = 0$  it equals 1 by construction; the test is whether a constant-MC cost function *fits*  $\hat{c}_{ms}$  (small, cost-shifter-driven  $\omega$ ), which is the substantive assumption.

Repeat 1–4 for cigarettes with  $A_{ms}^c$  if a cigarette cost function is needed for counterfactuals.

## 7. Solving counterfactuals

Prices are static (per-period) in each simulated month, so at each month of your forward simulation use the scenario's equilibrium price, then let the dynamic states (nicotine stock  $N_{im}$ , switching and adoption costs) evolve exactly as in your baseline.

**(a) Tax change  $\tau \rightarrow \tau'$  under  $\theta = 0$ .** Marginal cost is invariant, so  $p'_{ms} = \hat{c}_{ms} + \tau'_{ms} = p_{ms} + (\tau'_{ms} - \tau_{ms})$  — full pass-through, analytic, no fixed point. Feed  $p'_{ms}$  into the choice probabilities and re-simulate.

**(b) General  $\theta > 0$  (if you report it as a robustness case).** Price solves a fixed point because the markup depends on shares, which depend on price:

$$p_{ms}^{(t+1)} = \hat{c}_{ms} + \tau_{ms} + \frac{\theta}{|\beta^p| A_{ms}(p_e^{(t)}, p_c^{(t)})},$$

and symmetrically for  $p^c$ . Iterate both prices to convergence (a contraction for  $\theta$  not too large); then re-simulate demand. Under  $\theta = 0$  this reduces to (a).

**(c) E-cigarette ban.** Remove  $e$  from the choice set (nest Smoke =  $\{c\}$ ). Cigarette marginal cost is unchanged, so under  $\theta^c = 0$  the cigarette price is **unchanged** at  $\hat{c}_{ms}^c + \tau_{ms}^c$ ; recompute shares over  $\{c, \text{not smoke}\}$  and simulate. This is why your fixed-price ban result is robust under the pricing model the pass-through evidence supports. If instead  $\theta^c > 0$ , removing the substitute  $e$  raises cigarette demand and hence the cigarette markup/price via (b), which **attenuates** the induced switch to cigarettes — so your competitive result is an upper bound on the ban-induced increase in cigarette use, and the near-unit cigarette pass-through argues that the attenuation is small.

## 8. Assumptions and limitations

- **Constant marginal cost** in quantity within a market (supported by near-unit pass-through); rules out short-run capacity/upward-sloping supply.

- **Static (per-period) pricing:** no dynamic/forward-looking firm behavior, even though demand is dynamic; defensible for a competitive, low-margin index good.
- **Homogeneous good:** the representative e-cigarette is pod-weighted and disposables are undercovered in tracked retail (see companion note); the supply relation describes that tracked market, consistent with the household panel.
- **Conduct imposed, not tested:**  $\theta$  is set to the competitive value on the strength of external pass-through evidence rather than identified in-sample.